
Users
PK user_id INT
email VARCHAR(100)
password VARCHAR(100)
role VARCHAR(20)
status VARCHAR(20)
created_at TIMESTAMP
Customer
PK customer_id INT
FK user_id INT
name VARCHAR(100)
phone VARCHAR(15)
FOREIGN KEY (user_id) REFERENCES Users(user_id)
ServiceProvider
PK provider_id INT
FK user_id INT
experience_years INT
bio TEXT
verification_status VARCHAR(50)
FOREIGN KEY (user_id) REFERENCES Users(user_id)
Admin
PK admin_id INT
FK user_id INT
admin_role VARCHAR(50)
permissions TEXT
department VARCHAR(100)
FOREIGN KEY (user_id) REFERENCES Users(user_id)
Address
PK address_id INT
FK area_id INT
street VARCHAR(255)
landmark VARCHAR(255)
label VARCHAR(50)
latitude FLOAT
longitude FLOAT
FOREIGN KEY (area_id) REFERENCES Area(area_id)
Area
PK area_id INT
FK city_id INT
area_name VARCHAR(100)
pincode VARCHAR(10)
FOREIGN KEY (city_id) REFERENCES City(city_id)
City
PK city_id INT
city_name VARCHAR(100)
state VARCHAR(100)
Category
PK category_id INT
category_name VARCHAR(100)
description TEXT
Service
PK service_id INT
FK city_id INT
FK category_id INT
service_name VARCHAR(100)
description TEXT
base_price FLOAT
duration INT
is_active BOOLEAN
FOREIGN KEY (category_id) REFERENCES Category(category_id)
FOREIGN KEY (city_id) REFERENCES City(city_id)
ServiceVariant
PK variant_id INT
FK service_id INT
variant_name VARCHAR(100)
price FLOAT
duration INT
FOREIGN KEY (service_id) REFERENCES Service(service_id)
Booking
PK booking_id INT
FK coupon_id INT
FK address_id INT
FK provider_id INT
FK customer_id INT
scheduled_date DATE
scheduled_time TIME
total_amount FLOAT
special_instructions TEXT
created_at TIMESTAMP
FOREIGN KEY (customer_id) REFERENCES Customer(customer_id)
FOREIGN KEY (provider_id) REFERENCES ServiceProvider(provider_id)
FOREIGN KEY (address_id) REFERENCES Address(address_id)
FOREIGN KEY (coupon_id) REFERENCES Coupon(coupon_id)
Complaint
PK complaint_id INT
FK user_id INT
subject VARCHAR(100)
description TEXT
priority VARCHAR(20)
status VARCHAR(20)
resolution_notes TEXT
created_at TIMESTAMP
FOREIGN KEY (user_id) REFERENCES Users(user_id)
ProviderReview
PK review_id INT
FK provider_id INT
FK booking_id INT
rating FLOAT
comment TEXT
created_at TIMESTAMP
FOREIGN KEY (booking_id) REFERENCES Booking(booking_id)
FOREIGN KEY (provider_id) REFERENCES ServiceProvider(provider_id)
ServiceReview
PK review_id INT
FK service_id INT
FK booking_id INT
rating FLOAT
comment TEXT
created_at TIMESTAMP
FOREIGN KEY (booking_id) REFERENCES Booking(booking_id)
FOREIGN KEY (service_id) REFERENCES Service(service_id)
Payment
PK payment_id INT
FK booking_id INT
payment_method VARCHAR(50)
amount FLOAT
gateway_ref VARCHAR(100)
status VARCHAR(50)
paid_at TIMESTAMP
FOREIGN KEY (booking_id) REFERENCES Booking(booking_id)
Cancellation
PK cancel_id INT
FK booking_id INT
reason TEXT
refund_amount FLOAT
refund_status VARCHAR(50)
cancelled_at TIMESTAMP
FOREIGN KEY (booking_id) REFERENCES Booking(booking_id)
BookingStatusLog
PK log_id INT
FK booking_id INT
status VARCHAR(50)
remarks TEXT
timestamp TIMESTAMP
FOREIGN KEY (booking_id) REFERENCES Booking(booking_id)
Coupon
PK coupon_id INT
code VARCHAR(50) (Unique)
discount_type VARCHAR(50)
min_order FLOAT
discount_value FLOAT
usage_limit INT
valid_from DATE
valid_to DATE
BookingItem
PK item_no INT
FK booking_id INT
FK service_id INT
quantity INT
unit_price FLOAT
custom_price FLOAT
FOREIGN KEY (booking_id) REFERENCES Booking(booking_id)
FOREIGN KEY (service_id) REFERENCES Service(service_id)
ProviderAvailability
PK availability_id INT
FK provider_id INT
day_of_week VARCHAR(20)
start_time TIME
end_time TIME
FOREIGN KEY (provider_id) REFERENCES ServiceProvider(provider_id)
ProviderDocument
PK document_id INT
FK provider_id INT
document_type VARCHAR(50)
description TEXT
file_url TEXT
verification_status VARCHAR(50)
uploaded_at TIMESTAMP
FOREIGN KEY (provider_id) REFERENCES ServiceProvider(provider_id)

Minimal FD Set and Proof for BCNF:

1. Users

user_id → email
user_id → password
user_id → role
user_id → status
user_id → created_at

user_id is a candidate key so it is also a super key. Therefore, the left-hand side of the functional dependency is a superkey, and the relation satisfies BCNF.

2. Customer

customer_id → user_id
customer_id → name
customer_id → phone

customer_id is a candidate key so it is also a super key and hence the relation is in BCNF.

3. ServiceProvider

provider_id → user_id
provider_id → experience_years
provider_id → bio
provider_id → verification_status

provider_id is a candidate key so it is also a super key and hence the relation is in BCNF.

4. Admin

admin_id → user_id
admin_id → admin_role
admin_id → permissions
admin_id → department

admin_id is a candidate key so it is also a super key and hence the relation is in BCNF.

5. City

city_id → city_name
city_id → state

city_id is a candidate key so it is also a super key and hence the relation is in BCNF.

6. Area

area_id → city_id
area_id → area_name
area_id → pincode

area_id is a candidate key so it is also a super key and hence the relation is in BCNF.

7. Address

address_id → area_id
address_id → street
address_id → landmark
address_id → label
address_id → latitude
address_id → longitude

address_id is a candidate key so it is also a super key and hence the relation is in BCNF.

8. Category

category_id → category_name category_id → description

category_id is a candidate key so it is also a super key and hence the relation is in BCNF.

9. Service

service_id → category_id service_id → city_id service_id → service_name service_id → description service_id → base_price service_id → duration service_id → is_active

service_id is a candidate key so it is also a super key and hence the relation is in BCNF.

10. ServiceVariant

variant_id → service_id variant_id → variant_name variant_id → price variant_id → duration

variant_id is a candidate key so it is also a super key and hence the relation is in BCNF.

11. ProviderAvailability

availability_id → provider_id
availability_id → day_of_week
availability_id → start_time
availability_id → end_time

availability_id is a candidate key so it is also a super key and hence the relation is in BCNF.

12. ProviderDocument

document_id → provider_id
document_id → document_type
document_id → document_url
document_id → verification_status

document_id is a candidate key so it is also a super key and hence the relation is in BCNF.

13. Booking

booking_id → customer_id
booking_id → provider_id
booking_id → address_id
booking_id → coupon_id
booking_id → scheduled_date
booking_id → scheduled_time
booking_id → total_amount
booking_id → special_instructions
booking_id → created_at

booking_id is a candidate key so it is also a super key and hence the relation is in BCNF.

14. BookingItem

(booking_id, item_no) → service_id (booking_id, item_no) → quantity (booking_id, item_no) → unit_price (booking_id, item_no) → custom_price

(booking_id, item_no) is a candidate key so it is also a super key and hence the relation is in BCNF.

15. BookingStatusLog

log_id → booking_id log_id → status log_id → remarks log_id → timestamp

log_id is a candidate key so it is also a super key and hence the relation is in BCNF.

16. Payment

payment_id → booking_id payment_id → payment_method payment_id → amount payment_id → gateway_ref payment_id → status payment_id → paid_at

payment_id is a candidate key so it is also a super key and hence the relation is in BCNF.

17. Cancellation

cancel_id → booking_id cancel_id → reason cancel_id → refund_amount cancel_id → refund_status cancel_id → cancelled_at

cancel_id is a candidate key so it is also a super key and hence the relation is in BCNF.

18. Coupon

coupon_id → code coupon_id → discount_type coupon_id → min_order coupon_id → discount_value coupon_id → usage_limit coupon_id → valid_from coupon_id → valid_to

coupen_id is a candidate key so it is also a super key and hence the relation is in BCNF.

19. ProviderReview

review_id → provider_id review_id → booking_id review_id → rating review_id → comment review_id → created_at

review_id is a candidate key so it is also a super key and hence the relation is in BCNF.

20. ServiceReview

review_id → service_id review_id → booking_id review_id → rating review_id → comment review_id → created_at

review_id is a candidate key so it is also a super key and hence the relation is in BCNF.

21. Complaint

complaint_id → user_id complaint_id → subject complaint_id → description complaint_id → priority complaint_id → status complaint_id → resolution_notes complaint_id → created_at

complaint_id is a candidate key so it is also a super key and hence the relation is in BCNF.

DDL Script:

```
CREATE TABLE Users (
   user_id INT PRIMARY KEY,
   email VARCHAR(100) UNIQUE NOT NULL,
   password VARCHAR(100) NOT NULL,
   role VARCHAR(20),
   status VARCHAR(20),
   created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE Customer (
   customer_id INT PRIMARY KEY,
   user_id INT UNIQUE,
   name VARCHAR(100),
   phone VARCHAR(15),
   FOREIGN KEY (user_id) REFERENCES Users(user_id)
);

CREATE TABLE ServiceProvider (
   provider_id INT PRIMARY KEY,
   user_id INT UNIQUE,
   experience_years INT,
   bio TEXT,
   verification_status VARCHAR(50),
   FOREIGN KEY (user_id) REFERENCES Users(user_id)
);

CREATE TABLE Admin (
   admin_id INT PRIMARY KEY,
   user_id INT UNIQUE,
   admin_role VARCHAR(50),
   permissions TEXT,
   department VARCHAR(100),
   FOREIGN KEY (user_id) REFERENCES Users(user_id)
);

CREATE TABLE City (
   city_id INT PRIMARY KEY,
   city_name VARCHAR(100),
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```
    state VARCHAR(100)
);

CREATE TABLE Area (
    area_id INT PRIMARY KEY,
    city_id INT,
    area_name VARCHAR(100),
    pincode VARCHAR(10),
    FOREIGN KEY (city_id) REFERENCES City(city_id)
);

CREATE TABLE Address (
    address_id INT PRIMARY KEY,
    area_id INT,
    street VARCHAR(255),
    landmark VARCHAR(255),
    label VARCHAR(50),
    latitude FLOAT,
    longitude FLOAT,
    FOREIGN KEY (area_id) REFERENCES Area(area_id)
);

CREATE TABLE Category (
    category_id INT PRIMARY KEY,
    category_name VARCHAR(100),
    description TEXT
);

CREATE TABLE Service (
    service_id INT PRIMARY KEY,
    city_id INT,
    category_id INT,
    service_name VARCHAR(100),
    description TEXT,
    base_price FLOAT,
    duration INT,
    is_active BOOLEAN,
    FOREIGN KEY (city_id) REFERENCES City(city_id),
    FOREIGN KEY (category_id) REFERENCES Category(category_id)
);
```

```
CREATE TABLE ServiceVariant (
   variant_id INT PRIMARY KEY,
   service_id INT,
   variant_name VARCHAR(100),
   price FLOAT,
   duration INT,
   FOREIGN KEY (service_id) REFERENCES Service(service_id)
);

CREATE TABLE ProviderAvailability (
   availability_id INT PRIMARY KEY,
   provider_id INT,
   day_of_week VARCHAR(20),
   start_time TIME,
   end_time TIME,
   FOREIGN KEY (provider_id) REFERENCES ServiceProvider(provider_id)
);

CREATE TABLE ProviderDocument (
   document_id INT PRIMARY KEY,
   provider_id INT,
   document_type VARCHAR(50),
   description TEXT,
   file_url TEXT,
   verification_status VARCHAR(50),
   uploaded_at TIMESTAMP,
   FOREIGN KEY (provider_id) REFERENCES ServiceProvider(provider_id)
);

CREATE TABLE Coupon (
   coupon_id INT PRIMARY KEY,
   code VARCHAR(50) UNIQUE,
   discount_type VARCHAR(50),
   min_order FLOAT,
   discount_value FLOAT,
   usage_limit INT,
   valid_from DATE,
   valid_to DATE
);
```

```sql
CREATE TABLE Booking (
    booking_id INT PRIMARY KEY,
    customer_id INT,
    provider_id INT,
    address_id INT,
    coupon_id INT NULL,
    scheduled_date DATE,
    scheduled_time TIME,
    total_amount FLOAT,
    special_instructions TEXT,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (customer_id) REFERENCES Customer(customer_id),
    FOREIGN KEY (provider_id) REFERENCES ServiceProvider(provider_id),
    FOREIGN KEY (address_id) REFERENCES Address(address_id),
    FOREIGN KEY (coupon_id) REFERENCES Coupon(coupon_id)
);

CREATE TABLE BookingItem (
    booking_id INT,
    item_no INT,
    service_id INT,
    quantity INT,
    unit_price FLOAT,
    custom_price FLOAT,
    PRIMARY KEY (booking_id, item_no),
    FOREIGN KEY (booking_id) REFERENCES Booking(booking_id),
    FOREIGN KEY (service_id) REFERENCES Service(service_id)
);

CREATE TABLE BookingStatusLog (
    log_id INT PRIMARY KEY,
    booking_id INT,
    status VARCHAR(50),
    remarks TEXT,
    timestamp TIMESTAMP,
    FOREIGN KEY (booking_id) REFERENCES Booking(booking_id)
);

CREATE TABLE Payment (
```

```sql
    payment_id INT PRIMARY KEY,
    booking_id INT UNIQUE,
    payment_method VARCHAR(50),
    amount FLOAT,
    gateway_ref VARCHAR(100),
    status VARCHAR(50),
    paid_at TIMESTAMP,
    FOREIGN KEY (booking_id) REFERENCES Booking(booking_id)
);

CREATE TABLE Cancellation (
    cancel_id INT PRIMARY KEY,
    booking_id INT UNIQUE,
    reason TEXT,
    refund_amount FLOAT,
    refund_status VARCHAR(50),
    cancelled_at TIMESTAMP,
    FOREIGN KEY (booking_id) REFERENCES Booking(booking_id)
);

CREATE TABLE ProviderReview (
    review_id INT PRIMARY KEY,
    provider_id INT,
    booking_id INT,
    rating FLOAT,
    comment TEXT,
    created_at TIMESTAMP,
    FOREIGN KEY (provider_id) REFERENCES ServiceProvider(provider_id),
    FOREIGN KEY (booking_id) REFERENCES Booking(booking_id)
);

CREATE TABLE ServiceReview (
    review_id INT PRIMARY KEY,
    service_id INT,
    booking_id INT,
    rating FLOAT,
    comment TEXT,
    created_at TIMESTAMP,
    FOREIGN KEY (service_id) REFERENCES Service(service_id),
    FOREIGN KEY (booking_id) REFERENCES Booking(booking_id)
```

```sql
);

CREATE TABLE Complaint (
   complaint_id INT PRIMARY KEY,
   user_id INT,
   subject VARCHAR(100),
   description TEXT,
   priority VARCHAR(20),
   status VARCHAR(20),
   resolution_notes TEXT,
   created_at TIMESTAMP,
   FOREIGN KEY (user_id) REFERENCES Users(user_id)
);
```